Formula Sheet

HUJI 67594 · updated weekly

Every formula has a one-line meaning. Gold boxes list the mistakes in that week's slides and the correct version. The full explanations are in `weekNN.pdf`.

WEEK 01

Lecture 1: Introduction · Tutorial 1: Probability

Delay

- $d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$: the four delays at each hop.
- $d_{\text{trans}} = L/R$: push L bits onto a link of R bit/s. Depends on size and rate, not distance.
- $d_{\text{prop}} = d/s$: one bit travels distance d at speed s ($\approx 2 \times 10^8$ m/s in fibre $= \frac{2}{3}c$). Depends on distance, not size.
- Speed-of-light bound: Jerusalem–NY $9164/3 \times 10^5 \approx 30.6$ ms; in practice ≥ 70 ms one way.
- CPU cycles wasted per RTT $= f_{\text{CPU}} \times \text{RTT}$ (3 GHz $\times$ 140 ms $= 4.2 \times 10^8$).

Circuit switching

- Phases: establish (costs ≥ 1 RTT) $\rightarrow$ transfer $\rightarrow$ teardown. Resources reserved; unused slots are lost.
- TDM/FDM with n slots or bands: each circuit gets R/n. File time = setup $+ L/(R/n)$, independent of the number of links.
- Must reserve the peak P; utilization $= A/P$ (A = average rate). Voice $\approx 1/3$, bursty data $\leq 1\%$.

Packet switching

- Store-and-forward, N links, no queuing: one packet $N\frac{L}{R}$; K packets back to back $(N + K - 1)\frac{L}{R}$ (pipeline).
- Traffic intensity La/R (a = packets/s): must be < 1, otherwise the queue grows without bound; near 1 the delay is huge.
- Statistical multiplexing: N independent flows of mean A need $\approx NA$, not NP. Spread: std $\propto \sqrt{N}$, std/mean $\propto 1/\sqrt{N}$ (LLN). Fails if bursts are correlated.
- Users on a link: $X \sim \text{Bin}(n, q)$ active; overload probability $P(X > R/r)$ (35 users, $q = 0.1$, 10 fit: ≈ 0.0004).

Probability rules

- $P(A \mid B) = P(A \cap B)/P(B)$: B known, ask about A.
- Independent: $P(A \cap B) = P(A)P(B)$ (multiply). Disjoint: $P(A \cup B) = P(A) + P(B)$ (add). Disjoint $\neq$ independent.
- Union bound: $P(A \cup B) \leq P(A) + P(B)$; exactly: $P(A) + P(B) - P(A \cap B)$.
- Total probability: $P(D) = \sum_i P(D \mid B_i)P(B_i)$, $\{B_i\}$ a partition.
- Bayes: $P(B_j \mid D) = \dfrac{P(D \mid B_j)P(B_j)}{\sum_i P(D \mid B_i)P(B_i)}$.
- "At least one of n fails" $= 1 - (1 - p)^n$ (50 parts at 99%: 39.5%).

Random variables

- $\mathbb{E}X = \sum x\,P(X = x)$; $\mathbb{E}(\sum a_i X_i) = \sum a_i \mathbb{E}X_i$ always (even dependent). $\text{Var}\,X = \mathbb{E}X^2 - (\mathbb{E}X)^2$.
- Indicator: $\mathbb{E}[I_A] = P(A)$.

	$P(X = k)$ / density	$\mathbb{E}X$	$\text{Var}\,X$
$\text{Ber}(p)$	$P(1) = p$	p	$p(1-p)$
$\text{Bin}(n,p)$	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
$\text{Geo}(p)$	$(1-p)^{k-1}p,\ k \geq 1$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$\text{Pois}(\lambda)$	$e^{-\lambda}\lambda^k/k!$	λ	λ
$\text{Uni}(a,b)$	$\frac{1}{b-a}$ on $[a,b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$, $P(X > x) = e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$N(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2

- Geometric: *tries* until success, mean $1/p$; *failures* before success, mean $(1-p)/p$.
- $\text{Bin}(n,p) \approx \text{Pois}(np)$ when n is large and p small.
- Memoryless: $P(X > s + t \mid X > t) = P(X > s)$. Only Exp (continuous) and Geo (discrete).

Poisson process (rate λ)

- Independent increments; count in any interval of length t is $\text{Pois}(\lambda t)$; no bunching.
- Inter-arrival times X_i (time *since the previous* arrival) are i.i.d. $\text{Exp}(\lambda)$; epochs $S_i = X_1 + \cdots + X_i$.
- Exactly one arrival in T: $\lambda T e^{-\lambda T}$, maximal ($= 1/e$) at $\lambda T = 1$ (this is slotted ALOHA, week 2).

Worked patterns

- Chain of n links, each fails w.p. p: reach D w.p. $(1-p)^n$; end-to-end resend: $\mathbb{E}[\text{tries}] = 1/(1-p)^n$; hop-by-hop: $1/(1-p)$ per hop, $n/(1-p)$ in total.
- Good/bad packet survival e^{-t} / e^{-1000t}: $P(\text{surv}) = pe^{-t} + (1-p)e^{-1000t}$; $P(\text{good} \mid \text{surv})$ by Bayes.

SLIDE CORRECTIONS · WEEK 01

- `L1, Strowger` "invents the telephone system" $\Rightarrow$ he invented the *automatic switch*.
- `L1, buffers` "even with infinite buffers packets are lost" $\Rightarrow$ infinite buffers never drop; the delay grows without bound, so packets arrive too late.
- `T1, P` $P : X \rightarrow [0,1]$ on elements $\Rightarrow$ P is defined on *events* (subsets of Ω).
- `T1, conditional` "B occurs if we know A" for $P(A \mid B)$ $\Rightarrow$ B is known, we ask about A.
- `T1, Bayes` $P(A \mid D) = \frac{P(D|A)P(A)}{\sum_i P(D|B_i)P(B_i)}$ $\Rightarrow$ valid only if A is one of the B_i.

- T1, RV $\{\omega \subseteq \Omega\}$; biased coin $I_X = b \Rightarrow \omega \in \Omega$; $X(H) = 1$ with $P(X = 1) = b$.
- T1, chain links "disjoint", "union" = product $\Rightarrow$ links *independent*; the product is the *intersection*.
- T1, Gaussian Var $= \sigma \Rightarrow$ Var $= \sigma^2$.
- T1, Poisson X_i = time for the i-th packet $\Rightarrow$ time since the *previous* arrival.

WEEK 02 — Lecture 2: Layering & Broadcast · Recitation 2: ALOHA

Layers

- 5 layers (bottom up): Physical (bits) / Link (frames, local MAC addr.) / Network (packets, global addr., routing, only IP) / Transport (segments, processes; TCP, UDP) / Application (messages).
- Routers: layers 1–3. Hosts: all 5.
- OSI: 7 layers, adding Session (5) and Presentation (6). In the Internet, OSI 5–7 are all "application".
- Layer = service (what) + interface (how to access) + protocol (how peers talk). Protocol = syntax (format, order) + semantics (meaning, actions).
- Encapsulation: each layer adds a header. TCP 20 + IP 20 + Ethernet 14 = 54 B of headers (+4 B CRC trailer, +8 B preamble on the wire). Useful fraction $= D/(D + 54)$.
- Narrow waist: A apps, L links need $A + L$ adaptations instead of $A \times L$.

MAC: the basics

- Wish list, rate R: one node alone $\to R$; M busy $\to R/M$ each; decentralized; no clock sync; simple.
- TDMA/FDMA, N stations: each gets R/N *always*, even when alone. No collisions; idle slots are wasted.
- Goodput (efficiency) = fraction of time with a *successful* transmission. Throughput = fraction with *any* transmission (in [KR]: a rate in bit/s). Useful rate $= \eta B$.
- $T = L/B$: slot = time to send one frame of L bits at bandwidth B.

ALOHA, binomial (n nodes, each sends w.p. p)

- **Slotted**: one given node succeeds $p(1-p)^{n-1}$; some node succeeds (disjoint, add):

$$\eta = np(1-p)^{n-1}, \qquad p^* = \tfrac{1}{n},$$

$$\eta^* = (1 - \tfrac{1}{n})^{n-1} \to \tfrac{1}{e} \approx 0.37.$$

- **Pure**: vulnerable window $2T$, each other node must skip 2 slots:

$$\eta = np(1-p)^{2(n-1)}, \quad p^* = \tfrac{1}{2n-1}, \quad \eta^* \to \tfrac{1}{2e} \approx 0.18.$$

- $(1 - \frac{1}{N})^N \to \frac{1}{e}$ and $(1 - \frac{1}{N})^{N-1} \to \frac{1}{e}$.
- Slot types at $p = 1/n$, many nodes: 37% success, 37% empty, 26% collision.
- Finite-n optimum is above the limit: slotted n=2: 0.50, n=5: 0.41; pure n=2: 0.296, n=5: 0.216.

ALOHA, Poisson (G attempts per slot)

- Slotted: $\eta = Ge^{-G}$, max $1/e$ at $G = 1$. Empty slot e^{-G}; collision $1 - e^{-G} - Ge^{-G}$.
- Pure: $\eta = Ge^{-2G}$, max $1/(2e)$ at $G = \frac{1}{2}$.
- Binomial with $p = G/n$, $n \to \infty$ gives Poisson. If λ is per second, $G = \lambda T$.

Goodput: two views

- One slot: $\eta = \mathbb{E}[T_{\text{suc}}]/T = P_{\text{suc}}$.
- Wait for success: slots until success $\sim$ Geo(P_{suc}), so $\eta = T/(T/P_{\text{suc}}) = P_{\text{suc}}$. Time between successes $= T/P_{\text{suc}}$.

CSMA / CSMA/CD (Ethernet)

- CSMA: listen first. Collisions still happen because of the propagation delay. Plain CSMA wastes the whole frame.
- CD: abort when a collision is heard. Worst-case detection $\approx 2\,\text{PROP}$, so

$$\text{TRANSP} \geq 2\,\text{PROP} \iff L_{\min} = 2\,\text{PROP} \cdot R = \tfrac{2dR}{s}.$$

- Slot $= 2\,\text{PROP}$; won w.p. $\alpha = Np(1-p)^{N-1}$; wasted slots $A = \frac{1-\alpha}{\alpha}$ ($\alpha = 0.4 \Rightarrow A = 1.5$).
- $\eta = \dfrac{\text{TRANSP}}{\text{TRANSP} + A \cdot 2\,\text{PROP}} = \dfrac{1}{1+3a}$ (simple); $\approx \dfrac{1}{1+5a}$ (precise), $a = \dfrac{\text{PROP}}{\text{TRANSP}} = \dfrac{t_{\text{prop}}}{t_{\text{trans}}}$.
- Backoff after the m-th collision: $K \in \{0, \ldots, 2^{\min(m,10)} - 1\}$, wait $K \cdot 512$ bit times. Bit time $= 1/R$ (0.1 μs at 10 Mbps; 512 $\to$ 51.2 μs). Two nodes collide again w.p. $1/2^m$. Jam = 48 bits.

Addresses, WiFi

- MAC address: 48 bits, flat, portable; IEEE fixes 24 bits, the vendor 24 (2^{24} per block). IP: hierarchical, not portable.
- 802.11: no CD (own signal drowns others; hidden terminal; fading) $\Rightarrow$ CSMA/CA: wait DIFS; if busy, random backoff that counts down only while idle; receiver ACKs after SIFS; no ACK $\Rightarrow$ larger backoff.
- RTS/CTS: short RTS (may collide, cheaply); the CTS is heard by everyone, including hidden nodes $\Rightarrow$ no data-frame collisions.

Exam patterns (recitation)

- *One specific* node succeeds: $p \cdot$ (others silent), no factor n. *Any* node (network goodput): multiply by n, i.e. $\binom{n}{1}$.
- Choice of network or case: law of total probability, $pP_1 + (1-p)P_2$.
- Pure node vs. slotted node: each pure node must skip *every* own slot that overlaps. Vulnerable window = own length + other's length.
- Q1: C one shot $= q[p(1-q)^2 + (1-p)(1-q)^6]$; upper goodput $= p \cdot 3q(1-q)^2 + (1-p) \cdot 2q(1-q)$.
- Q2 (n pure nodes with slot T, n slotted with slot $2T$): slotted node $p(1-p)^{n-1}(1-p)^{3n}$; pure node $\frac{1}{2}p(1-p)^{2(n-1)}[(1-p)^n + (1-p)^{2n}]$.
- Q2c (all synchronized): with $s = np(1-p)^{n-1}$, $z = (1-p)^n$: $\eta = sz^2 + sz = sz(1+z)$. Linearity of expectation avoids case splits.

SLIDE CORRECTIONS · WEEK 02

- L2 s.19 `Content-Length: 21` ⇒ "Site under construction" is 23 bytes.
- L2 s.30 narrow "waste" ⇒ narrow *waist*.
- L2 s.42/96 "collision domain" = "broadcast domain" ⇒ same on a bus only; a switched star has one broadcast domain and one collision domain per cable.
- L2 s.45 partitioning is "centralized" ⇒ needs only an agreement in advance; [KR] has 3 classes: partitioning, random access, taking turns.
- L2 s.103 time/frequency "modulation" ⇒ time/frequency *division*.
- L2 s.67 success probability "drops" to $1-p$ ⇒ it *rises* (from $(1-p)^n$); $1/(1-p)$ is per hop, $n/(1-p)$ in total.
- L2 s.73 $Np(1-p)^{N-1}$ is "approx." ⇒ exact (disjoint events); only the limit $1/e$ is approximate.
- L2 s.84 $\alpha_{\max} \approx 37\% \approx 40\%$ ⇒ $(1-\frac{1}{N})^{N-1}$: 41% at N=5, 37% only as $N \to \infty$; with $\alpha = 1/e$, $A = e-1 \approx 1.72$.
- R2 s.6 Wi-Fi = layer 1; "MAC" = a layer-2 protocol ⇒ Wi-Fi and Ethernet are layers 1+2; MAC is a sub-layer (ALOHA, CSMA/CD, CSMA/CA).
- R2 s.19 rate λ "per T" with $\text{Poi}(\lambda T)$ ⇒ λ per second ⇒ $\text{Poi}(\lambda T)$; per slot ⇒ $\text{Poi}(\lambda)$.
- R2 s.20 4 models announced ⇒ only the binomial ones are done; Poisson results above.
- R2 s.30 $P(\cdot) \cap P(\cdot)$; $P(\text{some node sends}) = np$ ⇒ $P(A \cap B)$; $P(\text{some sends}) = 1-(1-p)^n$; the n comes from adding disjoint successes.
- R2 s.31 goodput "is" $1/(2e)$ ⇒ only as $n \to \infty$; larger for finite n.